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Overactive Bladder

W. Stuart Reynolds, and Joshua A. Cohn

Keywords

lower urinary tract symptoms; overactive bladder; urgency urinary incontinence; urinary frequency; urinary incontinence; urinary urgency

Questions

1. Which definition or definitions appear in the current International Continence Society (ICS) terminology (2002)?
    a. Detrusor hyperreflexia
    b. Overactive bladder
    c. Idiopathic detrusor overactivity
    d. Detrusor instability
    e. b and c
2. Which symptoms are included in overactive bladder syndrome (OAB)?
    a. Dysuria
    b. Straining
    c. Urgency incontinence
    d. Bladder pain
    e. Stress incontinence
3. In the community, what percentage of adults have overactive bladder symptoms?

a. Less than 5%
b. 5% to 10%
c. 10% to 20%
d. 20% to 50%
e. More than 50%

4. Mixed incontinence includes:
   a. stress urinary incontinence.
   b. continuous incontinence.
   c. postmicturition leakage.
   d. incontinence during sexual intercourse.
   e. giggle incontinence.
5. Which features are characteristics of urgency?
   a. A normal sensation.
   b. It builds up slowly.
   c. It is usually felt suprapubically.
   d. It develops quickly and may lead to incontinence.
   e. a and b.
6. Which statements are TRUE for detrusor overactivity (DO)?
   a. It is characterized by phasic involuntary detrusor contractions during bladder filling.
   b. It is always accompanied by a feeling of urgency.
   c. It is a urodynamic diagnosis.
   d. It is a feature of the voiding phase of micturition.
   e. a and c.
7. Detrusor overactivity can be diagnosed:
   a. only if involuntary filling phase contractions are greater than 15 cm H_2O in amplitude.
   b. if an involuntary contraction is seen during bladder filling, irrespective of size.
   c. if there is urgency incontinence but no contraction.
   d. if leakage occurs during exercise.
   e. before urodynamics if the patient has overactive bladder syndrome.
8. Which statement is FALSE with respect to the hypotheses of detrusor pathophysiology?

a. Afferent sensitization signifies an increased afferent firing rate in response to a standardized stimulus.
b. The myogenic and integrative hypotheses require abnormal propagation of excitation in the bladder wall.
c. Synaptic reorganization in the spinal cord may contribute to neurogenic detrusor overactivity.
d. Detrusor overactivity requires volitional control.
e. Synergic coordination of bladder and outlet are determined at the brainstem/midbrain level.

9. Which of the following statements is correct regarding symptom assessment tools?
    a. OAB can be diagnosed from the frequency volume chart.
    b. A validated questionnaire is mandatory for diagnosis of OAB.
    c. Modular components of the International Consultation on Incontinence Questionnaire system (ICIQ) have undergone formal validation.
    d. A high urgency score is diagnostic of detrusor overactivity.
    e. a and b.

Answers

1. **e. b and c.** Detrusor hyperreflexia has been replaced by neurogenic detrusor overactivity. The concept of tone is poorly understood; hence, the term *detrusor instability* is not recommended.
2. **c. Urgency incontinence.** Urgency incontinence is experienced by a large proportion of patients with OAB.
3. **c. 10% to 20%. Two large prevalence surveys in North America and Europe have shown the prevalence at 16%, whereas the EPIC study put the figure at nearly 12%.** Prevalence of at least one lower urinary tract symptom is greater than 50% according to the EPIC study.
4. **a. Stress urinary incontinence.** This is one of the two constituents of mixed incontinence, along with urgency urinary incontinence.
5. **d. It develops quickly and may lead to incontinence.** Urgency is thought to be a symptom of rapid onset. Urgency leads to urgency

incontinence in the susceptible patient, but not invariably.

6. **e. a and c.** Involuntary contractions seen on urodynamic assessment are characteristic of detrusor overactivity.
7. **b. If an involuntary contraction is seen during bladder filling, irrespective of size.**
8. **d. Detrusor overactivity requires volitional control.** Patients are unable to inhibit overactive contractions voluntarily.
9. **c. Modular components of the International Consultation on Incontinence Questionnaire system (ICIQ) have undergone formal validation.** The ICIQ generates new tools and adopts established tools for a system that allows selection of appropriate symptom assessment for specific clinical contexts.

Chapter Review

1. DO is a urodynamic diagnosis. OAB is a symptom-based diagnosis, defined as urgency with or without urgency incontinence, usually with frequency and nocturia with no proven infection or other obvious pathology.
2. The pattern of voided volumes in patients with OAB is erratic.
3. Urgency with at least one other symptom is essential to the diagnosis of OAB.
4. The etiology of DO has been hypothesized to be due to neurogenic or myogenic disorders. Neurogenic disorders may involve (1) reduced suprapontine inhibition, (2) overexpression of primitive spinal bladder reflexes, (3) synaptic plasticity in which new reflexes develop as a response to C-fiber afferent neurons, and (4) sensitization of peripheral afferent terminals. Myogenic disorders involve spontaneous excitation, which may be a result of upregulation of surface membrane receptors.
5. A frequency voiding chart or voiding diary is essential for assessing OAB.
6. The two main urodynamic findings with overactive bladder are DO and increasing filling sensation.

7. Treatment of OAB should begin with conservative management, including lifestyle changes, followed by pharmacotherapy and, finally, for intractable cases, surgical therapy that involves either nerve stimulation or surgical procedures on the bladder itself.
8. Aging, neurologic disease, female gender, bladder outlet obstruction, and the metabolic syndrome are potential contributors to OAB.

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The Underactive Detrusor

Christopher R. Chapple, and Nadir I. Osman

Keywords

bladder outlet resistance; detrusor underactivity; underactive detrusor

Questions

1. Which of the following is NOT part of the International Continence Society's (ICS's) definition of detrusor underactivity (DUA)?
    a. A contraction of reduced duration
    b. Prolonged bladder emptying
    c. Incomplete bladder emptying
    d. A contraction of reduced strength
    e. A contraction of reduced speed
2. A symptom syndrome of "underactive bladder" (UAB) is defined as:
    a. reduced desire to void, usually accompanied by urinary frequency or nocturia, with or without incontinence that predominates at night.
    b. reduced desire to void, associated with incomplete bladder emptying.
    c. symptoms of impaired bladder emptying in the absence of bladder outlet obstruction.

d. infrequent voiding, associated with voiding symptoms and increased postvoid residual.
e. There is currently no recognized definition for UAB.

3. What is the ultrastructural pattern associated with DUA, according to the classification of Elbadawi?
   a. Dense band pattern
   b. Degeneration pattern
   c. Dysfunction pattern
   d. Disjunction pattern
   e. Myelohypertrophy pattern
4. Most current diagnostic criteria estimate which aspect of detrusor contraction?
   a. Efficiency
   b. Sustainability
   c. Speed
   d. Strength
   e. Duration
5. Which of the following criteria is not thought to be affected by the presence of bladder outlet obstruction?
   a. Detrusor contraction duration
   b. Bladder contractility index
   c. Detrusor contraction coefficient
   d. Projected isovolumetric pressure
   e. Watt factor
6. Which of the following statements regarding parasympathomimetic agents in DUA is TRUE?
   a. Muscarinic agonists show good efficacy in restoring contractility.
   b. Muscarinic agonists are more likely to be effective in patients with complete bladder denervation.
   c. Muscarinic agonists can cause severe cardiac depression.
   d. Anticholinesterases can cause uterine contraction.
   e. None of the above.
7. What is the proposed mechanism of action of intravesical electrotherapy (IVE)?
   a. Direct stimulation of detrusor myocytes

b. Stimulation of mechanoreceptive afferent nerves
c. Direct stimulation of efferent nerves
d. Inhibition of pathologic urethral afferent signaling
e. None of the above

8. Which of the following features appears to be associated with more positive outcomes in the treatment of DUA?
   a. Neurogenic etiology
   b. Older age
   c. Intact bladder sensation
   d. Myogenic etiology
   e. Acontractile bladder
9. Which of these neurologic disorders is most frequently associated with DUA on urodynamics?
   a. Parkinson disease
   b. Multiple sclerosis
   c. Multisystem atrophy
   d. Cerebrovascular accident—postacute phase
   e. Brain tumor
10. What is the natural history of DUA in men without known neurogenic bladder dysfunction?
    a. Deterioration in symptoms but not urodynamic parameters
    b. No significant change in urodynamic parameters for at least 10 years
    c. Improvement in contractility in at least 50% of individuals
    d. Most will require bladder outlet surgery by 10 years
    e. No studies are available

Answers

1. **e. A contraction of reduced speed. The 2002 standardization report of the ICS defines DUA as "a contraction of reduced strength and/or duration, resulting in prolonged bladder emptying and/or failure to achieve complete bladder emptying within a normal time span."** An acontractile detrusor is separately

defined as “one that cannot be demonstrated to contract during urodynamic studies.”

2. **e. There is currently no recognized definition for UAB.** A symptom syndrome of “underactive bladder” is difficult to rationally define because of the absence of studies correlating individual symptoms to the underlying detrusor abnormality. Even then, empirical evidence would suggest that symptoms of DUA are very diverse and overlap significantly with those of overactive bladder (OAB).
3. **b. Degeneration pattern. The “degeneration” pattern is associated with DUA and consists of widespread disrupted detrusor myocytes and axonal degeneration.** Elbadawi proposed that distinct ultrastructural patterns observed by electron microscopy characterized the normally contractile aging detrusor and different bladder dysfunctions. Although the applicability of this classification system is disputed, other groups have noted similar findings.
4. **d. Strength. Most criteria focus on contraction strength as derived from detrusor pressure at maximal flow.**
5. **e. Watt factor.** The watt factor is a mathematical calculation and provides a measure of bladder power. Its major advantages are that it minimally depends on volume and is not affected by increased outlet resistance. However, it is a complex calculation with no validated cutoffs.
6. **c. Muscarinic agonists can cause severe cardiac depression.** Muscarinic agonists have been associated with reports of severe myocardial depression leading to cardiac arrest. This, and their lack of efficacy, has led to their nonuse in clinical practice.
7. **b. Stimulation of mechanoreceptive afferent nerves. IVE activates mechanosensitive bladder afferents (myelinated Aδ fibers) and restores bladder sensation. It is postulated that repeat activation of this pathway upregulates its performance during volitional voiding.**
8. **c. Intact bladder sensation.** Intact bladder sensation is necessary to normally trigger the micturition reflex. Its presence suggests some residual bladder innervation and was found to be associated with

better responses to some treatments of DUA (e.g., urethral sphincter botulinum neurotoxin A and muscarinic agonists)

9. **c. Multisystem atrophy.** Multisystem atrophy is a neurodegenerative disease that may be confused with Parkinson disease. It is associated with autonomic dysfunction and DUA in at least half of patients because of atrophy of parasympathetic nerves.
10. **b. No significant change in urodynamic parameters for at least 10 years.** There are very few data on the natural history of DUA. A longitudinal follow-up study (10 years) by Thomas et al. showed no significant symptomatic or urodynamic deterioration in men with DUA managed conservatively at 10 years.

Chapter Review

1. Detrusor underactivity (DUA) is a contraction of reduced strength and/or duration, resulting in prolonged bladder emptying and/or failure to achieve complete bladder emptying within a normal time span.
2. DUA often coexists with other lower urinary tract dysfunctions in the elderly.
3. Normal aging results in a reduction in autonomic innervation and a decline in sensory function in the lower urinary tract.
4. Diabetes mellitus impairs detrusor function through both myogenic and autonomic dysfunction.
5. Lumbosacral spinal cord trauma or disk disease and pelvic surgery can lead to injury of the pelvic plexus.
6. Afferent nerves play a central role in the initiation and maintenance of a detrusor contraction.
7. Treatment strategies to address DUA include timed voiding, double voiding, pelvic floor physiotherapy, and intermittent self-catheterization.
8. Intravesical electrotherapy activates mechanosensitive bladder afferents (myelinated Aδ fibers) and restores bladder sensation. It is postulated that repeat activation of this pathway upregulates its performance during volitional voiding.

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Nocturia

Stephen D. Marshall, and Jeffrey P. Weiss

Keywords

desmopressin; frequency volume chart; global polyuria; hyponatremia; nocturia; nocturnal bladder capacity index; nocturnal polyuria index; nocturnal polyuria; voiding diary

Questions

1. Nocturnal polyuria is defined as:
    a. nocturnal polyuria index (NPi) greater than 0.33.
    b. nocturnal urine volume (NUV) greater than 6.4 mL/kg.
    c. NUV greater than 54 mL/h.
    d. a and b.
    e. a, b, and c.
2. All of the following statements are true about nocturia EXCEPT:
    a. nocturia is voiding that is preceded and followed by sleep.
    b. the prevalence of nocturia increases with age.
    c. nocturia impairs sleep efficiency, sleep latency, and slow-wave sleep and is associated with increased mortality.
    d. one or more voids per night appear to be clinically significant.
    e. nocturia is associated with falls.
3. When instructing a patient how to complete a frequency-volume chart, it is essential to:
    a. tell the patient to record what time he or she goes to sleep.

b. tell the patient to record what time he or she awakens.
c. tell the patient to record how much he or she drinks during the day.
d. a and b.
e. a, b, and c.

4. Factors that inhibit antidiuretic hormone (ADH) secretion include all of the following EXCEPT:
   a. hyperkalemia.
   b. atrial natriuretic peptide (ANP).
   c. hypercalcemia.
   d. prostaglandin E2 (PGE2).
   e. lithium.
5. A 75-year-old obese man (100 kg) with a short neck reports frequent urination during the nighttime and completes a voiding diary. His 24-hour voided volume is 2000 mL, and his NUV is 1000 mL. Based on these diary findings, this man has:
   a. global polyuria.
   b. nocturnal polyuria.
   c. diminished nocturnal bladder capacity.
   d. diminished global bladder capacity.
   e. none of the above.
6. According to a recent study, which of the following showed the greatest decline in nocturia severity and greatest improvement of health-related quality of life (HRQL) in men with benign prostatic hyperplasia (BPH) and nocturia?
   a. Watchful waiting.
   b. α-Blockers.
   c. Transurethral resection of the prostate (TURP).
   d. Transurethral microwave treatment (TUMT).
   e. There was no difference among these treatments.
7. When bound to V2 receptors in the renal collecting tubules, desmopressin:
   a. increases water permeability.
   b. enhances water reabsorption.
   c. dilutes extracellular fluid.
   d. concentrates urine.

e. all of the above.

8. Current thinking is that desmopressin is most appropriate to treat:
    a. nocturnal polyuria.
    b. global polyuria.
    c. decreased global bladder capacity.
    d. decreased nocturnal bladder capacity.
    e. all of the above.
9. Which of the following have caused statistically but minimally clinically significant reductions in nocturia episodes?
    a. α-Blockers
    b. 5α-Reductase inhibitors
    c. Antimuscarinics
    d. α-Blockers and antimuscarinics
    e. All of the above
10. Regarding the frequency volume chart (voiding diary), which of the following statements is FALSE?
    a. NUV is defined as the total volume of all voids preceded and followed by sleep.
    b. The first morning void counts toward total number of daily voiding episodes.
    c. Nighttime may be during daylight hours.
    d. Maximum voided volume is the single greatest urine volume measured during a 24-hour period.
    e. Nocturia or enuresis occurs when the nocturia index (Ni) is greater than 1.
11. When prescribing desmopressin to an elderly patient (>65 years), it is advisable to:
    a. check a baseline serum sodium level
    b. check a serum sodium level within 7 days and then 28 days after initial or incremental dosing
    c. check a serum sodium level every 6 months or as often as indicated
    d. avoid prescribing in those with baseline serum sodium below 130 mmol/L
    e. all of the above

Answers

1. **e. a, b, and c.** Nocturnal polyuria is defined as NPi greater than 0.20 in young adults and greater than 0.33 in adults older than 65 years when 24-hour urine production is within normal limits. Other definitions include NUV greater than 0.9 mL/min (54 mL/h) and NUV greater than 1.5 mL/min (90 mL/h). A universally accepted definition of nocturnal polyuria has yet to be identified.
2. **c. One or more voids per night appear to be clinically significant.** Several studies have shown that two or more voids per night generate bother and impair quality of life.
3. **d. a and b.** Although having the patient record how much he drinks during the day may be helpful, it is essential that the patient record the time he goes to sleep and the time he awakens to know the number of nocturnal voids and NUV.
4. **a. Hyperkalemia. Factors that inhibit antidiuretic hormone (ADH) and cause diuresis (inhibit water reabsorption) include PGE2, ANP, hypercalcemia, hypokalemia, lithium, and tetracyclines.** Reversal of water diuresis, accordingly, may occur through stimulation of V2 receptors, either by endogenous arginine vasopressin or a congener thereof, such as desmopressin.
5. **b. Nocturnal polyuria. This patient suffers from nocturnal polyuria as defined by his NPi. His NPi can be calculated by dividing his NUV by his 24-hour voided volume:**

$$\begin{aligned} \text{NPi} &= \text{NUV} / 24-\text{hour voided volume} \\ &= 1000\ \text{mL} / 2000\ \text{mL} = 0.5 \end{aligned}$$

He has nocturnal polyuria because his NPi is greater than 0.33. Global polyuria is defined as 24-hour urine volume greater than 40 mL/kg. Diminished nocturnal bladder capacity and global

bladder capacity cannot be determined by the information provided in this scenario.

6. **c. Transurethral resection of the prostate (TURP).** After 6 to 12 months, watchful waiting, α-blockers, TURP, and TUMT yielded reduction in nocturia episodes by 7%, 17%, 75%, and 32%, respectively. Improvements in HRQL were most strongly associated with treatment-associated declines in nocturia severity.
7. **e. All of the above.** When bound to V2 receptors in the renal collecting tubules, desmopressin increases water permeability, enhances water reabsorption, dilutes extracellular fluid, and concentrates urine.
8. **a. Nocturnal polyuria.** Current thinking is that desmopressin would be the most appropriate therapy for patients with nocturia related to nocturnal polyuria.
9. **e. All of the above.** α-blockers, 5α-reductase inhibitors, antimuscarinics, and antimuscarinics + α-blockers have been found to produce a statistically significant reduction in nocturia episodes, yielding minimal clinical significance.
10. **a. NUV is defined as the total volume of all voids preceded and followed by sleep.** This is incorrect. NUV is the total volume of urine passed during the night and the first morning void.
11. **e. All of the above.** When prescribing desmopressin to a patient older than 65 years of age, it is advisable to closely monitor serum sodium AND to avoid prescribing it to elderly patients with baseline hyponatremia.

Chapter Review

1. There is a clear impact of aging on the prevalence of nocturia. Younger people are more likely to manifest decreased nocturnal bladder capacity, whereas older people manifest overproduction of urine.
2. Accumulation of fluid in the dependent parts of the body (third spacing) and return of fluid to the circulating volume when the patient is recumbent may be an underlying cause of nocturia.

3. The pathophysiology of nocturia may be related to increased mean arterial blood pressure when supine and an alteration of circadian rhythm of ADH (ordinarily there is an increased production of ADH at night during the hours of sleep).
4. Twenty percent of people aged 20–40 years awake to void two or more times compared with 60% of those 70 years or older.
5. Two or more voids a night are clinically significant.
6. Obstructive sleep apnea is a common cause of nocturia.
7. Desmopressin may result in hyponatremia; women are more sensitive to the drug than are men.

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Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

Karl-Erik Andersson, and Alan J. Wein

Keywords

bladder; emptying and voiding discussion; lower urinary tract

Questions

(Multiple answers are possible throughout.)

1. The effects of administration of an antimuscarinic agent to an individual with an overactive bladder (OAB) include all of the following EXCEPT:
    a. increased total bladder capacity.
    b. depressed amplitude of involuntary bladder contractions.
    c. increased outlet resistance.
    d. increased volume to the first involuntary bladder contraction.
    e. increased mean volume voided.
2. Which of the following muscarinic receptor subtypes is the most common in human detrusor smooth muscle?
    a. M_1
    b. M_2
    c. M_3

d. M_4
e. M_5

3. Which of the following muscarinic receptor subtypes is predominantly responsible for the mediation of bladder contraction in human detrusor smooth muscle?
   a. M_1
   b. M_2
   c. M_3
   d. M_4
   e. M_5
4. The use of antimuscarinic agents to treat OAB is limited by their lack of uroselectivity. Which of the following is NOT a recognized side effect of antimuscarinic agents?
   a. Dry mouth
   b. Constipation
   c. Cognitive dysfunction
   d. Bradycardia
   e. Blurred vision
5. Which of the following characteristics increases the possibility for an antimuscarinic agent to pass the blood-brain barrier?
   a. High lipophilicity
   b. Large molecular size
   c. Low electrical charge
   d. Quaternary ammonium structure
   e. Small molecular size
6. Anticholinergics exert their favorable effects on OAB by affecting:
   a. peripheral afferent (sensory) transmission.
   b. ganglionic transmission.
   c. central neural afferent transmission.
   d. muscarinic receptors on the detrusor smooth muscle.
   e. nicotinic receptors on the detrusor smooth muscle.
7. The Committee on Pharmacologic Treatment of the Fifth (2013) International Consultation on Incontinence assessed agents according to the Oxford Guideline, according to level of evidence

and grade of recommendation with respect to treatment of detrusor overactivity (DO). Which of the following (multiple answers are possible) did NOT receive a level of evidence rating of 1 and a grade of recommendation of A?

a. Fesoterodine
b. Flavoxate
c. Trospium
d. Solifenacin
e. Oxybutynin

8. Which of the following is NOT a class effect of antimuscarinics?

a. Accommodation paralysis
b. Constipation
c. Increased heart rate
d. Prolongation of the QT interval
e. Dry mouth

9. Of the following choices, which are correct? The bladder contains how many β-receptor subtypes (answer 1) _____ with the (answer 2) _____ having the most important functional role.

a. one; β_2
b. two; β_2
c. three; β_3
d. four; β_2
e. five; β_5

10. Which of the following pharmacologic actions is most likely responsible for the effects of oxybutynin when given systemically?

a. Antimuscarinic, direct muscle relaxant, and local anesthetic actions, equally
b. Direct muscle relaxant effect alone
c. Direct muscle relaxant effect and local anesthetic action
d. Antimuscarinic and direct muscle relaxant effects
e. Antimuscarinic effect

11. Which of the following muscarinic receptor subtypes are NOT known to be involved in the potential antimuscarinic side effects of dry mouth, constipation, tachycardia, drowsiness, and blurred vision?

a. M_1
b. M_2
c. M_3
d. M_4
e. M_5

12. The primary adverse event reported with the use of oxybutynin-transdermal has been:
    a. tachycardia.
    b. dry mouth.
    c. constipation.
    d. application site reactions.
    e. blurred vision.
13. Which of the following agents is relatively selective for M_3 receptor blockade?
    a. Darifenacin
    b. Oxybutynin
    c. Solifenacin
    d. Tolterodine
    e. Trospium
14. Mirabegron probably exerts its favorable effect on OAB by:
    a. activating the β_3 adrenergic receptor.
    b. competitive blockade of the β_3 receptor.
    c. inhibiting filling-induced activity in bladder afferent nerves.
    d. activating the α-adrenergic receptor.
    e. inhibiting the α-adrenergic receptor.
15. Of the following agents, which is actively excreted by the kidney in the proximal convoluted tubules?
    a. Darifenacin
    b. Oxybutynin
    c. Solifenacin
    d. Tolterodine
    e. Trospium
16. Which is (are) TRUE about tadalafil?

a. It improves International Index of Erectile Function (IIEF) scores.
b. It improves International Prostate Symptom Score (IPSS) scores.
c. It improves peak flow rates.
d. It is approved for the treatment of lower urinary tract symptoms (LUTS) due to benign prostatic obstruction (BPO).
e. It decreases detrusor pressure at peak flow.

17. Which of the following is NOT listed as a common side effect of imipramine?
    a. Systemic antimuscarinic effects
    b. Weakness, fatigue
    c. Priapism
    d. Cardiac arrhythmia
    e. Hepatic dysfunction
18. In men with BPO and OAB, adding an antimuscarinic to an α-adrenergic blocking agent will:
    a. significantly increase residual urine volume.
    b. significantly decrease detrusor pressure at peak flow.
    c. significantly decrease peak and mean flow rate.
    d. cause a, b, and c.
    e. none of the above.
19. Intravesical botulinum toxin subtype A:
    a. activates synaptosomal-associated protein (SNAP) 25 and the soluble N-ethylmaleimide-sensitive factor attachment *protein* (SNARE) complex.
    b. acts only on smooth muscle.
    c. inhibits the peripheral release of acetylcholine.
    d. is effective only in patients with both OAB and DO.
    e. reduces firing from bladder afferents.
20. Vaginal estrogen is effective in the treatment of:
    a. urogenital atrophy.
    b. stress urinary incontinence (SUI).
    c. urgency urinary incontinence.
    d. mixed urinary incontinence.
    e. anterior vaginal prolapse.

21. The side effects of the alpha-adrenergic agonists include all of the following EXCEPT:
    a. tremor.
    b. palpitations.
    c. hypertension.
    d. somnolence.
    e. respiratory difficulties.
22. Which of the following has been reported to increase stroke risk in women younger than 50 years?
    a. Phenylpropanolamine
    b. Ephedrine
    c. Pseudoephedrine
    d. Midodrine
    e. Clenbuterol
23. In theory, which of the following agents, from the standpoint of potential efficacy and safety, would be preferred for the treatment of SUI in a hypertensive individual?
    a. Ephedrine
    b. Propranolol
    c. Phenylpropanolamine
    d. Pseudoephedrine
    e. Clenbuterol
24. Which of the following statements is NOT true with respect to duloxetine hydrochloride?
    a. It significantly increases sphincteric muscle activity during filling/storage in an animal model.
    b. It is a serotonin-norepinephrine reuptake inhibitor.
    c. It is lipophilic and well absorbed.
    d. It is effective in decreasing SUI episodes in women.
    e. It is not metabolized by the liver.
25. Oral estrogen with or without progesterones in postmenopausal women:
    a. improves mixed urinary incontinence.
    b. worsens the risk of incontinence.
    c. worsens urinary incontinence in women with incontinence.
    d. improves SUI.

e. improves urgency incontinence.

26. A 75-year-old man placed on desmopressin therapy 3 days previously presents with a change of mental status and mental confusion. The most likely cause is:
    a. cognitive dysfunction due to antimuscarinic effect.
    b. hyponatremia.
    c. hypernatremia.
    d. hypokalemia.
    e. hyperkalemia.
27. With regard to bethanechol chloride, the least objective evidence exists to support which of the following statements?
    a. It has relatively selective in vitro action on urinary bladder and bowel.
    b. It has little or no nicotinic action.
    c. It is cholinesterase resistant.
    d. It causes in vitro contraction of bladder and smooth muscle.
    e. It facilitates bladder emptying.
28. What oral dose of bethanechol chloride is required to produce the same urodynamic effects, at least in a denervated bladder, at the subcutaneous dose of 5 mg?
    a. 200 mg
    b. 100 mg
    c. 50 mg
    d. 25 mg
    e. 10 mg
29. Prostaglandins have been hypothesized to affect bladder activity through all of the following actions EXCEPT:
    a. neuromodulation of efferent and afferent neurotransmission.
    b. sensitization to sensory stimuli (activation occurs with a lower degree of filling).
    c. activation of certain sensory nerves.
    d. potentiation of acetylcholine release from cholinergic nerve terminals.
    e. potentiation of adenosine triphosphate release from bladder mucosa.

30. With respect to α- and β-adrenergic receptors in the lower urinary tract and prostate, all the following are true EXCEPT:
    a. α receptors outnumber β receptors in the smooth muscle of the bladder base and proximal urethra.
    b. β receptors outnumber α receptors in the bladder body.
    c. α_1 receptors are more common than α_2.
    d. Lower urinary tract and prostate adrenergically induced smooth muscle contraction is mediated largely by α_1 receptors.
    e. Proximal urethral smooth muscle contraction is mediated primarily by α_1 receptors.
31. Which of the following α-adrenergic blocking agents has significant antagonistic properties at both α_1 and α_2 receptor sites?
    a. Prazosin
    b. Terazosin
    c. Phenoxybenzamine
    d. Doxazosin
    e. Tamsulosin
32. Available data suggest that which of the following side effects is more common with tamsulosin than with either terazosin or doxazosin?
    a. Dizziness
    b. Asthenia
    c. Postural hypotension
    d. Palpitations
    e. Retrograde ejaculation
33. Which of the following agents or classes of agents, when administered systemically, will selectively relax the striated musculature of the pelvic floor?
    a. Benzodiazepines
    b. Dantrolene
    c. Baclofen
    d. Botulinum toxin
    e. None of the above
34. Which of the following is the most widely distributed inhibitory neurotransmitter in the mammalian central nervous system?

a. γ-Aminobutyric acid (GABA)
b. Glycine
c. Glutamate
d. Dopamine
e. Norepinephrine

35. Baclofen (Lioresal) acts to decrease striated sphincter activity by which of the following mechanisms?
    a. Facilitating neuronal hyperpolarization through the $GABA_A$ receptor
    b. Activating the $GABA_B$ receptor and depressing monosynaptic and polysynaptic excitation of motor neurons and interneurons in the spinal cord
    c. Inhibiting excitation-contraction coupling in skeletal muscle by decreasing calcium release from the sarcoplasmic reticulum
    d. Inhibiting excitation-contraction coupling by preventing calcium entry into the cell
    e. Inhibiting acetylcholine release at the neuromuscular junction
36. All of the following statements are true with regard to botulinum toxin EXCEPT:
    a. It has been reported to be useful in the treatment of striated sphincter dyssynergia via direct sphincteric injection.
    b. It has been reported to be useful in the treatment of DO by direct intradetrusor injection.
    c. It inhibits the release of acetylcholine and other transmitters at the neuromuscular junction of somatic nerve and striated muscle and the autonomic nerves in smooth muscle.
    d. It has been reported to be of use, via periurethral striated muscle injections, in the treatment of SUI.
    e. The immunologic subtype utilized for urologic use has primarily been type A.
37. All of the following statements are true regarding the action of atropine and atropine-like agents (antimuscarinic agents) in patients with OAB and DO EXCEPT:
    a. Volume to first involuntary contraction increases.
    b. Total bladder capacity increases.

c. Heart rate decreases.
d. Urgency episodes decrease.
e. Amplitude of the DO contractions decreases.

38. Which of the following statements is FALSE with respect to atropine resistance?
    a. It is secondary to release of norepinephrine from pelvic nerve in addition to acetylcholine.
    b. It is of little importance in normal human detrusor function.
    c. Its importance in treatment of DO in humans remains to be established.
    d. It applies to the response of the whole bladder to pelvic nerve stimulation, but not to the response of the detrusor to exogenous cholinergic stimulation.
    e. It is commonly invoked as a cause for only partial clinical improvement in the treatment of OAB with antimuscarinic agents.
39. Which statement is FALSE regarding M_3 receptors?
    a. They are less common than M_2 receptors in detrusor smooth muscle.
    b. They are more common than M_2 receptors in urothelium.
    c. They are the most important muscarinic receptor for detrusor contraction.
    d. They are blocked by atropine.
    e. When activated, they lead to an increase in intracellular calcium in detrusor smooth muscle cells.
40. All of the following are well-known potential adverse events of antimuscarinic therapy EXCEPT:
    a. constipation.
    b. cognitive dysfunction.
    c. increased heart rate.
    d. blurred vision.
    e. hyperhidrosis.
41. Match the side effects with the predominate muscarinic receptor subtype.
    1. M1

2. M2
3. M3
4. M4
5 M5
6. none

a. constipation
b. dry mouth
c. cognitive dysfunction
d. increased heart rate
e. increased QT interval

42. When used in the usual dosages for the treatment of OAB in a patient who is not on clean intermittent catheterization, antimuscarinic agents act primarily by (more than one response may be correct):
    a. reducing detrusor voiding contraction.
    b. decreasing activity in C fibers.
    c. decreasing activity in Aδ fibers.
    d. reducing the micromotions caused by the release of small packets of acetylcholine.
    e. reducing excitation of afferent nerves from the urothelium and detrusor.
43. An 80-year-old man has OAB-wet and cognitive dysfunction. Theoretically, the antimuscarinic drug that would be expected to be the safest with respect to worsening his cognition is:
    a. solifenacin.
    b. oxybutynin ER.
    c. fesoterodine.
    d. trospium.
    e. tolterodine.
44. Activation of detrusor smooth muscle by both acetylcholine and adenosine triphosphate requires:
    a. increase in intracellular potassium concentration.
    b. decrease in intracellular potassium concentration.
    c. increase in intracellular calcium concentration.
    d. increase in intracellular cyclic guanosine monophosphate (GMP).

e. increase in intracellular cyclic adenosine monophosphate (AMP).

45. Which of the following have 1A or 1B ratings (modified Oxford system) for treatment of DO?
    a. Flavoxate
    b. Dicyclomine
    c. Estrogen
    d. Tamsulosin
    e. Tolterodine
    f. Fesoterodine
    g. Darifenacin
    h. Solifenacin
    i. Propiverine
    j. Oxybutynin
    k. Trospium
46. Which of the following drugs can produce or aggravate SUI in a woman?
    a. Alfuzosin
    b. Nifedipine
    c. Tamsulosin
    d. Oxybutynin
    e. Propantheline
    f. Flavoxate
    g. Fesoterodine
    h. Duloxetine
47. Intravesical DMSO is:
    a. generally used in a 70% solution.
    b. generally used in a 50% solution.
    c. useful for the treatment of neurogenic DO.
    d. useful for the treatment of bladder pain syndrome (interstitial cystitis).
    e. useful for the treatment of idiopathic DO.
48. Regarding the vanilloids, which of the following is/are TRUE?
    a. They act primarily to render C fibers insensitive.
    b. They act primarily to render Aδ fibers insensitive.

c. When delivered intravesically, they cause a biphasic (excitation then blockade) effect.
d. Resiniferatoxin is much more potent than capsaicin for desensitization but proportionately less so for excitation.
e. Capsaicin is more potent than resiniferatoxin for desensitization.

49. A 30-year-old paraplegic man is wet between intermittent catheterizations (catheterizes five times per day). He is on solifenacin, 10 mg daily, and reports moderate dry mouth and increased difficulty in his bowel regimen. A reasonable next step in treatment is to:
    a. increase dose of solifenacin.
    b. add oxybutynin ER, 10 mg daily.
    c. add darifenacin, 7.5 mg daily.
    d. add darifenacin, 15 mg daily.
    e. use intradetrusor botulinum toxin.
50. Generally, treatment with intradetrusor botulinum toxin:
    a. must be repeated every 3 to 12 months.
    b. improves quality of life in patients incontinent due to neurogenic DO.
    c. loses efficacy with repeat treatments.
    d. can cause urinary retention.
    e. requires general anesthesia.
51. The Heart and Estrogen/Progestin Replacement Study (HERS), Women's Health Initiative (WHI), and Nurses' Health Study compositely showed that:
    a. oral estrogen plus progesterone worsened urinary incontinence in older postmenopausal women with incontinence.
    b. oral estrogen plus progesterone increased the incidence of SUI and urgency urinary incontinence in those continent at baseline.
    c. oral estrogen and progesterone worsened the frequency of incontinence in those incontinent at baseline.
    d. transvaginal estrogen improves SUI in postmenopausal women.

e. the risk of developing incontinence was increased in postmenopausal women taking estrogen alone or estrogen with progestin.

52. Adrenergically induced smooth muscle contraction in the human lower urinary tract is mediated primarily by which receptor?
    a. α_{1D}
    b. β_3
    c. β_2
    d. α_{1A}
    e. α_2

Answers

1. c**. Increased outlet resistance. Atropine and atropine-like agents will depress normal bladder contractions and involuntary bladder contractions of any cause.** In such patients, the volume to the first involuntary bladder contraction will generally be increased, the amplitude of the involuntary bladder contraction decreased, and the total bladder capacity increased. Outlet resistance, at least as reflected by urethral pressure measurements, does not seem to be clinically affected.
2. **b. M_2 .** On the basis of existing knowledge, it is now recommended that the designations M_1 to M_5 be used to describe both the pharmacologic subtypes and the molecular subtypes of muscarinic acetylcholine receptors. **The human urinary bladder smooth muscle contains a mixed population of M_2 and M_3 subtypes, with M_2 receptors predominant (M_2 receptors predominate at least 3:1 versus M_3 receptors not only on detrusor cells but also on other bladder structures, which may be of importance for detrusor activation).**
3. **c. M_3 . The minor population of M_3 receptors is generally accepted at this time as primarily responsible for the mediation of bladder contraction.**

4. **d. Bradycardia.** In general, drug therapy for lower urinary tract dysfunction is hindered by a concept that can be expressed in one word: uroselectivity. **The clinical utility of available antimuscarinic agents is limited by their lack of selectivity, responsible for the classic peripheral antimuscarinic side effects of dry mouth, constipation, blurred vision, tachycardia, and effects on cognitive function.**
5. **a, c, and e.** High lipophilicity, small molecular size, and low electrical charge increase the possibilities for an antimuscarinic agent to pass the blood-brain barrier. **Quaternary ammonium compounds pass into the central nervous system to a limited extent. Tertiary amines pass the blood-brain barrier to a greater extent.**
6. **a and d.** The traditional view was that in OAB/DO, antimuscarinics act by blocking the muscarinic receptors on the detrusor muscle that are stimulated by acetylcholine released from the activated cholinergic (parasympathetic) nerves. However, antimuscarinic drugs act mainly during the storage phase of micturition, decreasing urgency and increasing bladder capacity, and during this phase there is normally no parasympathetic input to the lower urinary tract. **There is good experimental evidence that antimuscarinics decrease the activity in both C and Aδ afferent fibers during the filling/storage phase of micturition.**
7. **b. Flavoxate.** A level of 1 implies the presence of systematic reviews, meta-analysis, and good-quality randomized controlled clinical trials. A grade of recommendation of A means that the agent is highly recommended based on level 1 evidence. Flavoxate received a level of evidence of 2, meaning that either randomized control trials and/or good-quality prospective cohort studies existed regarding its use, but a grade of recommendation of D, meaning that no recommendation for use is possible because of inconsistent/inconclusive evidence.
8. **d. Prolongation of the QT interval.** Well-known peripheral antimuscarinic side effects include blurred vision due to accommodation paralysis, constipation due to impaired bowel

motility, increase in heart rate due to some blockade of M_2 cardiac receptors, and dry mouth due primarily to blockade of M_3 receptors in the salivary glands. QT prolongation is not related to muscarinic blockade but rather linked to inhibition of the hERG potassium channel in the heart. Some antimuscarinic drugs may in fact cause this, but this is not a class effect.

9. **c. Three; β_3.** Three cloned subtypes of β-adrenergic receptors, β_1, β_2, β_3, have been identified in the detrusor of most species, including humans. Studies have revealed a predominant expression of the β_3 receptor, and there is functional evidence for an important role in both normal and neurogenic bladders. The β_3 agonist mirabegron represents the first drug in this class to be developed for treatment of OAB.
10. **e. Antimuscarinic effect.** Oxybutynin has several pharmacologic effects, some of which seem difficult to relate to its effectiveness in the treatment of DO. It has antimuscarinic, direct muscle relaxant, and local anesthetic actions. The local anesthetic action and direct muscle relaxant effect may be of importance when the drug is administered intravesically but probably play no role when it is given orally. In vitro, oxybutynin was shown to be 500 times weaker as a smooth muscle relaxant than as an antimuscarinic agent. Most probably, when given systematically, oxybutynin acts mainly as an antimuscarinic drug.
11. **d and e.** The M_3 receptor has a primary role in salivation, bowel motility, and visual accommodation. The M_1 receptor is thought to be involved in cognition. The M_2 receptor is the primary cholinergic receptor in the heart, causing bradycardia when activated and, potentially, tachycardia, when blocked. The M_4 and M_5 receptors do not at this time seem to have a primary role in any of these organ systems.
12. **d. Application site reactions.** The transdermal delivery of oxybutynin alters oxybutynin metabolism, reducing production of the primary metabolite, responsible for most of the side effects, to an even greater extent than extended-release oxybutynin. The

primary adverse event for this preparation has been application-site reaction: pruritus in 14% and erythema in 8.3%.

13. **a. Darifenacin.** Darifenacin is relatively selective for M_3 receptor blockade, meaning that, in vitro, the affinity for M_3 receptors is greater than for the other muscarinic receptors. This is only a relative selectivity, however, and whether this translates into either greater efficacy or greater tolerability has yet to be established.
14. **a and c. Mirabegron is a β_3 agonist, the first commercially available and approved drug of its type, approved for the treatment of OAB.** It acts not only by stimulating the B_3 receptors in the bladder (causing detrusor relaxation) but also by inhibiting filling-induced activity in both mechanosensitive Aδ and C-fiber primary bladder afferents, at least in an animal model.
15. **e. Trospium.** Darifenacin, oxybutynin, solifenacin, and tolterodine are all actively metabolized in the liver by the cytochrome P450 enzyme system. Trospium chloride is not metabolized to any significant degree in the liver. It is actively excreted by the proximal convoluted tubules in the kidney.
16. **a, b, and d.** Tadalafil is a phosphodiesterase (PDE)-5 inhibitor. Phosphodiesterase inhibitors enhance the presumed cyclic AMP and cyclic GMP relaxation of lower urinary tract smooth muscles as well as blood vessels in the penis. It was originally developed for the treatment of erectile dysfunction and improved IIEF scores. The observation that patients treated for erectile dysfunction with PDE-5 inhibitors had an improvement of their lower urinary tract symptoms led to interest in use of these drugs to treat LUTS and OAB. **PDE-5 inhibitors significantly improve IPSS scores but do not improve peak flow rates compared with placebo.** Similarly, they do not change detrusor pressure at peak flow. The mechanism behind the beneficial effect of these substances on LUTS/OAB and their site(s) of action largely remains to be elucidated. They are effective, however, and in fact tadalafil, at the time of this writing, has been approved for the treatment of LUTS due to benign prostatic obstruction.

17. **c. Priapism. The most frequent side effects of the tricyclic antidepressants are those attributable to their systemic antimuscarinic activity.** Allergic phenomena (including rash), hepatic dysfunction, obstructive jaundice, and agranulocytosis may also occur, but rarely. Central nervous system side effects may include weakness, fatigue, parkinsonian effect, fine tremor noted most in the upper extremities, manic or schizophrenic picture, and sedation, probably from an antihistaminic effect. Postural hypotension may also be seen, presumably on the basis of selective blockade (a paradoxic effect) of α_1-adrenergic receptors in some vascular smooth muscle. Tricyclic antidepressants can also cause excess sweating of obscure cause and a delay of orgasm or orgasmic impotence, the cause of which is likewise unclear. They can also produce arrhythmias and interact in deleterious ways with other drugs, and so caution must be observed in their use in patients with cardiac disease. They have not been reported to cause priapism.
18. **e. None of the above. Several randomized control trials have demonstrated that the combination treatment of antimuscarinic drugs and α_1-adrenergic receptor antagonist is more effective at reducing male lower urinary tract symptoms than α-blockers alone in men with BPO and OAB.** Used in the recommended doses, and in men whose voiding was not significantly compromised previously, none of the parameters related to emptying changed appreciably (detrusor pressures, flow rates, residual voided volume).
19. **c and e. Botulinum toxin-A cleaves SNAP 25 and renders the SNARE complex inactive as its primary mechanism of action. It acts on both striated muscle and smooth muscle and in fact was first studied in striated muscle. It blocks acetylcholine release and reduces firing from bladder afferents.** It is effective in both neurogenic and idiopathic DO and also in patients with OAB. Successful OAB treatment does not appear to be related to the existence of DO.

20. **a. Urogenital atrophy. The evidence supporting the use of estrogens in lower urinary tract dysfunction remains controversial, but considerable data support the use of vaginal estrogen in urogenital atrophy. The vaginal route improves dryness, pruritus, and dyspareunia and provides a greater improvement in physical findings than oral administration.**
21. **d. Somnolence.** Potential side effects of all of these agents include blood pressure elevation, anxiety, and insomnia due to stimulation of the central nervous system; headache; tremor; weakness; palpations; cardiac arrhythmias; and respiratory difficulties. They should be used with caution in patients with hypertension, cardiovascular disease, or hyperthyroidism.
22. **a. Phenylpropanolamine.** The risk of hemorrhagic stroke in women younger than 50 years has been reported to be 16 times higher in those who have been taking phenylpropanolamine, an α-adrenergic agonist, as an appetite suppressant, and 3 times higher in women who have been taking the drug for less than 24 hours as a cold remedy, although the latter was not statistically significant (the former was). Phenylpropanolamine has been removed from the market in the United States.
23. **b. Propranolol.** Theoretically, β-adrenergic blocking agents, such as propranolol, might be expected to "unmask" or potentiate an α-adrenergic effect, thereby increasing urethral resistance. Such treatment has been suggested as an alternative treatment to α-adrenergic agonists in patients with sphincteric incontinence and hypertension. Some studies support such usage, but they are not randomized or controlled. The other compounds listed are α-adrenergic agonists and are a risk factor for increased blood pressure.
24. **e. It is not metabolized by the liver.** Duloxetine hydrochloride is a serotonin-norepinephrine reuptake inhibitor that has been shown, in an animal model, to significantly increase urethral sphincteric muscle activity during the filling/storage phase of micturition. It is lipophilic, well absorbed, and extensively metabolized by the liver. It is still approved as a treatment for SUI in some countries. It was withdrawn from the FDA approval process in the United States but

is licensed in the European Union for the treatment of SUI in women with moderate to severe incontinence, defined as 15 or more episodes per week.

25. **b and c. The results of the HERS study, the WHI study, and the Nurses' Health Study all suggest that there is no evidence that estrogens with or without progesterone should be used in the treatment of urinary incontinence. In fact, estrogen with or without progesterone increases the risk of urinary incontinence among continent postmenopausal women and worsens urinary incontinence in those already with incontinence.**
26. **b. Hyponatremia. Side effects are relatively uncommon during desmopressin treatment, but there is a risk of water retention and hyponatremia. It is recommended that serum sodium concentration be measured in elderly patients before and after a few days of treatment.**
27. **e. It facilitates bladder emptying. Many acetylcholine-like drugs exist, but only bethanechol chloride (Urecholine, Duvoid, others) exhibits a relatively selective in vitro action on the urinary bladder and gut with little or no nicotinic action.** Bethanechol chloride is cholinesterase resistant and causes an in vitro contraction of smooth muscle from all areas of the bladder. Although it has been reported to increase gastrointestinal motility and has been used in the treatment of gastroesophageal reflux, and although anecdotal success in specific patients with voiding dysfunction seems to occur, there is no evidence to support its success in facilitating bladder emptying in a series of patients when the drug was the only variable.
28. **a. 200 mg.** It is generally agreed that, at least in a "denervated" bladder, an oral dose of 200 mg is required to produce the same urodynamic effects as a subcutaneous dose of 5 mg.
29. **e. Potentiation of adenosine triphosphate release from bladder mucosa.** Prostanoids are synthesized both locally in bladder muscle and mucosa, with synthesis initiated by various physiologic stimuli such as detrusor muscle stretch, mucosal injury, and neural stimulation; directly by adenosine triphosphate; and by mediators of inflammation. Prostanoids have been variably reported to be

useful in facilitating bladder emptying with intravesical administration. Possible roles include (1) neuromodulators of efferent and afferent transmission; (2) sensitization; (3) activation of certain sensory nerves; and (4) potentiation of acetylcholine (but not ATP) release from cholinergic nerve terminals through prejunctional prostanoid receptors.

30. **c. α_1 Receptors are more common than α_2. The human lower urinary tract contains more α_2 than α_1 receptors, but adrenergically induced human lower urinary tract smooth muscle contraction and prostate smooth muscle contraction are mediated largely, if not exclusively, by α_1 adrenergic receptors.** The remainder of the statements are true.
31. **c. Phenoxybenzamine.** Phenoxybenzamine (Dibenzyline) was the α-adrenolytic agent originally used for the treatment of voiding dysfunction. It and phentolamine have blocking properties at both α-adrenergic receptor sites. Prazosin hydrochloride (Minipress) was the first potent selective α-adrenergic antagonist used to lower outlet resistance. **Terazosin (Hytrin) and doxazosin (Cardura) are two highly selective postsynaptic α_1-adrenergic blockers. Most recently, alfuzosin and tamsulosin (Flomax), both highly selective α_1-adrenergic blockers, have appeared and are marketed solely for the treatment of benign prostatic hyperplasia because of some reports suggesting preferential action on prostatic rather than vascular smooth muscle.**
32. **e. Retrograde ejaculation. Available data suggest that retrograde ejaculation and rhinitis are more common with tamsulosin, whereas dizziness and asthenia are more common with terazosin and doxazosin.**
33. **e. None of the above.** There is no class of pharmacologic agents that will selectively relax the striated musculature of the pelvic floor. Botulinum toxin-A when injected directly into the striated sphincter will relax it, but this "relative selectivity" is because of where it is locally injected.
34. **a. γ-Aminobutyric acid (GABA). GABA and glycine have been identified as major inhibitory transmitters in the central**

nervous system. GABA is the most widely distributed inhibitory neurotransmitter in the mammalian central nervous system. It appears to mediate the inhibitory actions of local interneurons in the brain and presynaptic inhibition within the spinal cord.

35. **b. Activating the GABA $_B$ receptor and depressing monosynaptic and polysynaptic excitation of motor neurons and interneurons in the spinal cord.** Benzodiazepines potentiate the action of GABA by facilitating neuronal hyperpolarization through the $GABA_A$ receptor. Baclofen (Lioresal) depresses monosynaptic and polysynaptic excitation of motor neurons and interneurons in the spinal cord by activating $GABA_B$ receptors. Dantrolene (Dantrium) exerts its effects by a direct peripheral action on skeletal muscle. It is thought to inhibit the excitation-induced release of calcium ions from the sarcoplasmic reticulum of striated muscle fibers, thereby inhibiting excitation-contraction coupling and diminishing the mechanical force of contraction. Botulinum A toxin (Botox) is an inhibitor of acetylcholine release at the neuromuscular junction of somatic nerves on striated muscle.
36. **d. It has been reported to be of use, via periurethral striated muscle injections, in the treatment of SUI.** Intrasphincteric injection of botulinum toxin A was first reported useful in the treatment of striated sphincteric dyssynergia in 1990. The toxin blocks the release of acetylcholine and other transmitters from presynaptic nerve endings by interacting with the protein complex necessary for docking vesicles. This results in decreased muscle contractility and muscle atrophy at the injection site. The drug has been reported to be of use as well in the treatment of neurogenic DO and cases of non-neurogenic DO. There are seven immunologically distinct antigenic subtypes. Types A and B are in clinical use in urology, but most studies and treatments have been carried out with botulinum toxin type A (Botox). Intrasphincteric injections of botulinum toxin are not useful for SUI. In fact, they can cause SUI in females.
37. **c. Heart rate decreases.** Those with an M_2 receptor blockade profile can increase heart rate, but the clinical significance of this is

unknown.

38. **a. It is secondary to release of norepinephrine from pelvic nerve in addition to acetylcholine.** The most common neurotransmitter mentioned as the prime alternate in atropine resistance is adenosine triphosphate. Norepinephrine is released by postganglionic sympathetic nerves (e.g., hypogastric), not by parasympathetic ones (e.g., pelvic).
39. **b. They are more common than M_2 receptors in urothelium.** M_2 receptors outnumber M_3 in urothelium as well as detrusor smooth muscle. All other statements are true.
40. **e. Hyperhidrosis.** If anything, antimuscarinic agents can cause decreased sweating. All the others are antimuscarinic side effects.
41. **a: 3; b: 3; c: 1; d: 2; e: 6.** Increased QT interval, caused by some antimuscarinic compounds, is not an antimuscarinic property. All the others are antimuscarinic and are caused primarily by blockade of the indicated receptor subtype.
42. **b, c, d, and e.** At the dose employed for OAB treatment, antimuscarinic agents do not cause significant reduction in the voiding contraction of voluntary micturition. They have been implicated in all the others. C and Aδ fibers refer to afferent nerves carrying noxious and "normal" stimuli, respectively.
43. **d. Trospium.** Trospium, as a quaternary amine, is lipophobic and does not penetrate well through the blood-brain barrier. The others, all tertiary amines, do pass the blood-brain barrier to a greater extent.
44. **c. Increase in intracellular calcium concentration.** This occurs through both extracellular influx and mobilization of intracellular calcium.
45. **e, f, g, h, i, j, and k.** Propiverine is a drug with "mixed action" and currently is not available in the United States, but is in Europe.
46. **a and c.** The α-adrenergic antagonists can decrease outlet resistance and thereby irritate or worsen SUI. Nifedipine is a calcium antagonist; fesoterodine and propantheline are anticholinergic agents; duloxetine is a serotonin-norepinephrine reuptake inhibitor; oxybutynin is primarily an antimuscarinic agent with some direct

smooth muscle relaxant effects on the bladder; and flavoxate has mixed actions and questionable effects.

47. **b and d.** DMSO is a naturally occurring compound with multiple pharmacologic actions used in a 50% solution for the treatment of bladder pain syndrome (including interstitial cystitis). It is not useful for the treatment of DO.
48. **a, c, and d.** It is possible that vanilloids have some effects on Aδ fibers as well. They cause an initial excitation followed by a long-lasting blockade. Resiniferatoxin is 1000 times more potent than capsaicin for desensitization and a few hundred times more potent for excitation.
49. **e. Use intradetrusor botulinum toxin.** Adding or increasing antimuscarinic medication will simply increase the severity of the adverse events already experienced. Intradetrusor botulinum toxin has been shown to be very effective in neurogenic DO and should be considered if following failure of, or intolerance to, antimuscarinic therapy.
50. **a, b, and d.** Repeat injections (two to nine) have not lost efficacy over time. The injections can be done without general anesthesia (local). Intravesical botulinum toxin A is effective in reducing neurogenic DO and does improve quality of life in such patients.
51. **a, b, c, and e. a** is from the HERS, **b** and **c** from the WHI, and e from the Nurses' Health Study. Although many clinicians prescribe transvaginal estrogen or estrogen plus progestin cream for symptoms of OAB or/and SUI, there is no real evidence that estrogen, with or without progesterone, is useful in the treatment of urinary incontinence.
52. **d.** α_{1A}**.** More α_2 than α_1 receptors are present, but adrenergically induced contraction is mediated largely by the α_{1A} (and in the detrusor, α_{1D}). β receptors cause smooth muscle relaxation. The β_3 subtype is the predominant β receptor in the human detrusor.

Chapter Review

1. The major neurohumoral stimulus for physiologic bladder contraction is acetylcholine-induced stimulation of postganglionic parasympathetic muscarinic cholinergic receptor sites in the bladder.
2. The muscarinic receptor functions may be changed in different urologic disorders without an overt neurogenic cause.
3. Behavioral therapy should always be used in conjunction with drug therapy for overactive bladder (OAB).
4. Calcium channel antagonists are not effective in treating OAB.
5. Potassium channel openers available today are not effective in treating OAB.
6. The use of estrogens to treat stress urinary incontinence (SUI) has resulted in worsening of preexisting urinary incontinence in patients with SUI and urgency urinary incontinence and new-onset incontinence in patients who have not had it. The use of vaginal estrogen in the treatment of urogenital atrophy improves dryness, pruritus, dyspareunia, and physical findings. The vaginal route is more effective than oral administration.
7. Intravesical oxybutynin has shown some efficacy in treating OAB as well as in treating intestinal augmented OABs.
8. Antimuscarinics may have effects on cognitive function, particularly in the elderly.
9. Antimuscarinics may be divided into tertiary and quaternary amines. The latter are not well absorbed and have limited ability to enter the central nervous system, unlike the tertiary amines.
10. Antimuscarinics may have adverse cardiac effects.
11. α_1-adrenergic blockers are not effective in women.
12. The human urinary bladder smooth muscle contains a mixed population of M_2 and M_3 subtypes, with M_2 receptors predominant (M_2 receptors predominant at least 3:1 versus M_3). M_3 receptors are primarily responsible for the mediation of bladder contraction.
13. Mirabegron is a β_3 agonist, the first commercially available drug of its type approved for the treatment of overactive bladder.
14. PDE-5 inhibitors significantly improve International Prostate Symptom Score scores but do not improve peak flow rates when compared with placebo.

15. Urinary tract smooth muscle contraction and prostate smooth muscle contraction are mediated largely, if not exclusively, by α_1-adrenergic receptors.